Tour de Maths Leg 2
King David High School Linksfield
19 February 2015

Q1

The number of employees who work at TdM Pty Ltd increased by 16% in 2013. In 2014, there was a further increase of 25%. What was the overall percentage increase of the number of workers over this 2 year period?

5

Q2

The rectangle ABCD shown below is rotated 360° about side BC. What is the volume of the resulting solid to the nearest cm^3?

5

Q3

Look at the following number "pattern" and then write down the next row:

1
11
21
1211
111221
312211
....

5

Q4

The maximum possible number of rectangular blocks with dimensions 3,5cm x 5cm x 4cm are placed in a larger rectangular box with dimensions 16cm x 14cm x 15cm. What is the total surface area of all the smaller blocks which are in the box?

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Q5

A bottle containing 200 tablets has a mass of 180g. The same bottle containing 150 tablets has a mass of 165g. What is the mass of the empty bottle (in grams)?

5

Q6

How many different triangles can be made with seven matches so that the perimeter of each triangle is the same as the total length of the seven matches

5

Q7

A dog is chained to the corner of a building on a 3m long leash. What area will the dog have in which to move? (Give the answer to the nearest m^2)

5

Q8

The next number in the pattern

2
3
6
15
42

is ....

5

Q9 Fifteen children stand in a circle and pass a ball in a clockwise direction around the circle. The ball is passed to every sixth player.
How many children never touch the ball?

5

Q10 Twice the number of marbles in bag A is less than the number of marbles in bag B. The sum of the number of marbles in bags A and C is less than the number of marbles in bag B. There are more marbles in bag D than there are in bag B. Bag C contains 6 marbles and there are 9 in bag D. How many marbles are there in bag B?

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Q11 The area of an equilateral triangle is $\frac{2}{3}$ the area of a regular hexagon (i.e. all 6 sides are the same length).

If the perimeter of the hexagon is 6, what is the **perimeter** of the triangle?

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Q12 Calvin was having an argument with Hobbes about whether Calvin could cook or not. After a while Hobbes said "Wow, wow, so cook." Calvin realised that this was a cryptogram as follows:

$$\begin{array}{r} WOW \\ WOW \\ \underline{SO} \\ \underline{COOK} \end{array}$$

If each letter represents a different digit and K = 9, find the value of W + C .

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Q13 What is the **units** digit in the sum $(5 + 1)(5^2 + 1)(5^3 + 1)\ldots(5^{2015} + 1)$?

10

Q14 A; B; C; D; E and F are 6 numbers. The average of A; B; C and D is 10; andt the average of B; C; D; E; and F is 14. F is twice the value of A. What is the average of A and E?

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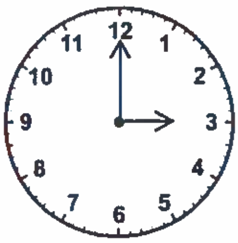
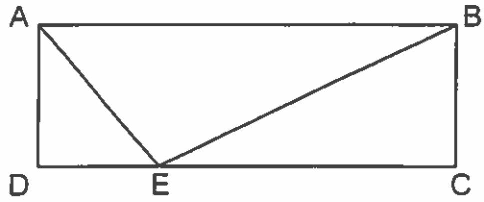
Q15 How many times in one 24 – hour day do the hands of a clock form a right angle?



10

Q16



AB = 5; AE = 3 and BE = 4. What is the area of ABCD?

20

Q17

If the eggs in a basket are removed two at a time, one egg remains. If the eggs are removed three at a time, two eggs remain. If the eggs are removed four, five or six at a time. then, three, four and five eggs respectively will remain in the basket. But, if they are taken out seven at a time, no eggs will be left over. What is the least number of eggs which could be in the basket?

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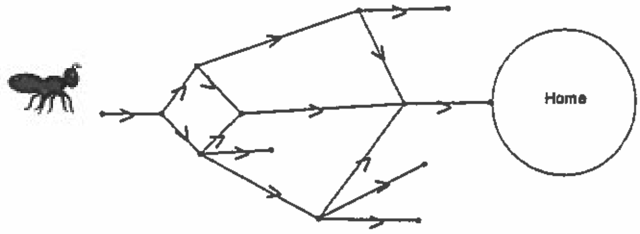
Q18



The ant is travelling home on the network of tracks. He cannot turn back. When he reaches a junction, each path ahead is an equally probable choice for him to take. What is the probability that he will reach home?

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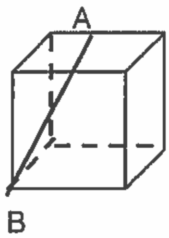
Q19



The diagram shows cube with sides of length 2cm. A is the midpoint of one of the sides. AB is joined. What is the length of AB?

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Q20

Four points P; Q; R and S are 1 metre apart, in that order, along a straight line. What is the shortest path, in metres, from P to S; if there must be at least 1 metre between Q and R at all times? Please give your answer correct to 2 decimal places.

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